

SNOWFLAKE

Cloud Data Platform for Swiss SMEs

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Agenda

01

Scenario

SwissBike SA challenges

02

Why Snowflake?

SME pain points solved

03

Architecture

Independent scaling &
query performance

04

Storage

Columnar, partitioned &
protected

05

Workflow

Data flow end to end

06

Vendor Lock-in

Multi-cloud, but at what
cost?

07

Cost Scenario

Real numbers for a Swiss
SME

08

Pros & Cons

Strengths & trade-offs

09

Market Alternatives

Snowflake vs BigQuery,
Redshift, Synapse

10

Recommendation

When should a SME use
Snowflake?

11

Demo

SwissBike SA use case

12

Reference

Further studies

Scenario / Challenges

SWISSBIKE SA

SwissBike SA is a Swiss SME selling bicycles and accessories online:

- ~50 employees, has a small engineering team
- ~1,500 orders/month
- currently 10 analytics users
- Collect data across disconnected systems(E-commerce, ERP, CRM and Marketing tools)

CHALLENGES

- Data is scattered across **multiple systems**
- Analytics rely on **manual Excel exports**
- **More users** need access to analytics
- **Growing data volume** with no scalable storage strategy
- **Budget constraints**
- **Query performance** degrades as data grows

Key Question

As the data engineering team, will Snowflake help us **centralize** data and **scale** analytics while staying **within budget**?

What Does Snowflake Solve?

SME PAIN POINT	SNOWFLAKE SOLUTION	COVERED IN
User increase	Add Virtual Warehouses without touching storage	Architecture
Data volume growing, query speed slower	Storage layer scales independently + Micro-partitions + pruning skip irrelevant data	Architecture + Storage
Multiple disconnected systems, no single source of truth	Snowpipe centralizes all sources	Workflow + Vendor lockin
Manual CSV exports, reports stale	Automated ingestion, always up to date	Workflow + Vendor lockin
Budget sensitive	Pay only when warehouse is running	Cost Scenario

Bottom Line

As data engineers, Snowflake removes cluster management overhead and let the team **focus on data products, not infrastructure.**

Architecture : Independent Scaling & Query Performance

Hybrid Model = Shared-disk + Shared-nothing + MPP

CLOUD SERVICES

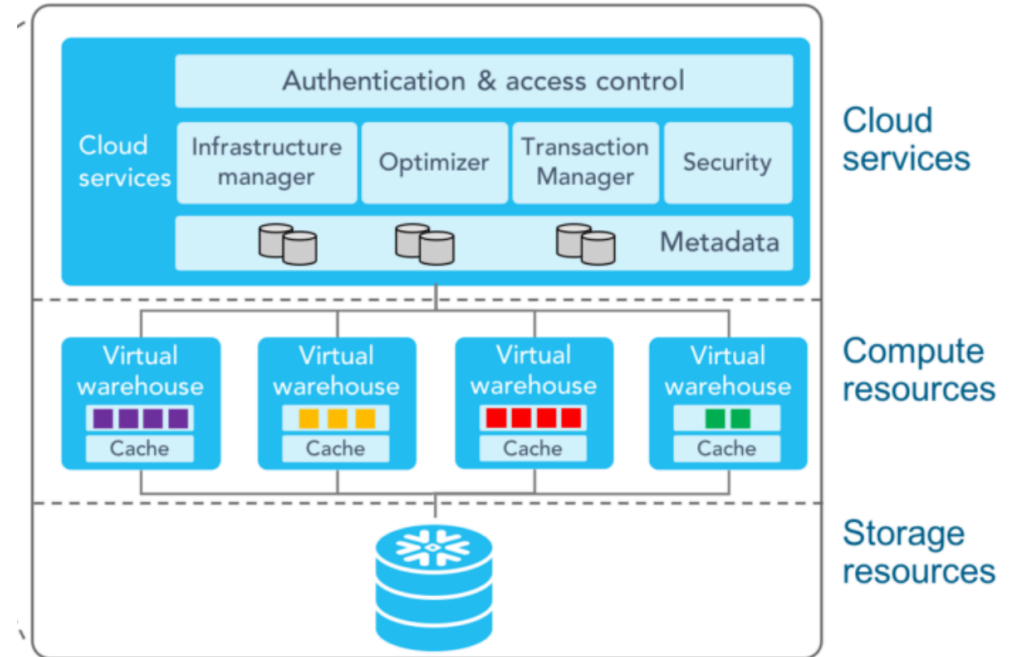
- Always on, managed by Snowflake (Query opt, metadata, auth)
- 24h query result **cache** → free repeated queries
- **Free** under 10% of daily compute usage

COMPUTE (VIRTUAL WAREHOUSES)

- Local disk cache
- **Auto scale** up/down, pause when idle
- Trade off : Cold start 2-3s

STORAGE

- Decoupled from compute → pay separately
- Trade off : Egress costs when different regions



Key insight:

Scale independently + Query without conflict, no duplication

Trade off:

Layer separation adds overhead for small frequent transactions (`INSERT`, `UPDATE` slower than PostgreSQL)

Storage: Columnar, Partitioned & Protected

DATA TYPES

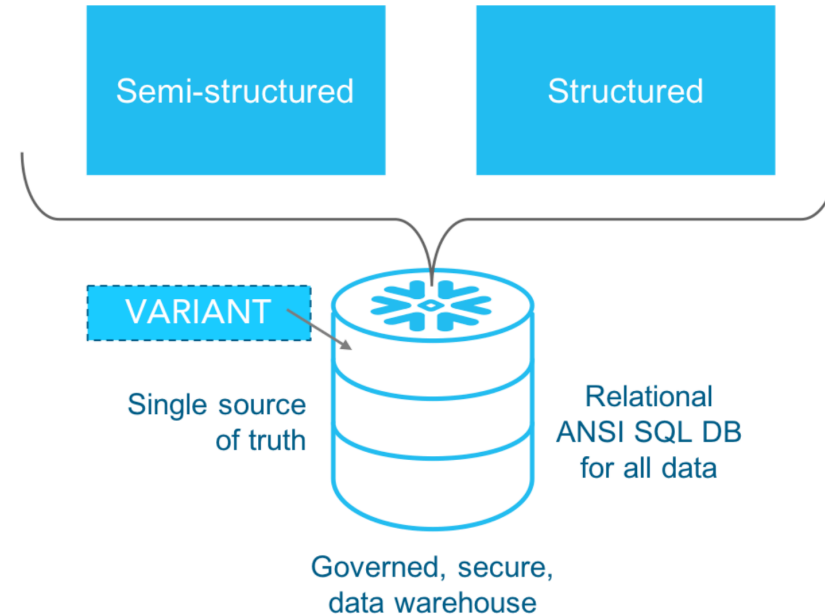
- **Structured + Semi-structured**
- **Unstructured** → PDF, images via staged storage — limited query support

MICRO-PARTITIONS

- Auto-split into **columnar chunks** (50–500MB)
- Metadata per partition → **pruning skips irrelevant data at query time**
- **Clustering keys** — optional, co-locates frequent query columns for large tables

DATA PROTECTION

- **Time Travel** — query or restore data up to 90 days
- **Fail-safe** — 7-day recovery after Time Travel expires, managed by Snowflake



Key Advantage

- Maintain query speed as data grow → prune+cluster
- Storage cost stays low → columnar compression

Trade-off

Egress costs apply when extracting data out → keep compute and storage in same region

Workflow

SwissBike SA centralizes data from multiple business systems into Snowflake to generate business insights.

DATA SOURCES

- E-commerce platform
- ERP system
- CRM system
- Marketing tools
- CSV / API imports

DATA INGESTION

- Batch imports
- Connectors
- External stages
- Semi-structured formats (JSON, Parquet)

SNOWFLAKE PROCESSING

- Centralized storage
- Automatic compression
- Micro-partitioning
- Query optimization
- MPP execution via Virtual Warehouses

OUTPUTS

- SQL analytics
- Dashboards (Power BI, Tableau)
- Reports
- Machine learning datasets

Vendor Lock-in: Multi-cloud, But At What Cost?

PORTABLE

- Runs on AWS, Azure, GCP : no vendor dependency
- Standard ANSI SQL : queries portable
- Open ingestion formats — Parquet, JSON, CSV

PROPRIETARY FEATURES

FEATURE	LOCK-IN RISK	MITIGATION
Time Travel	no equivalent elsewhere	Use for recovery only
Zero-Copy Clone	proprietary syntax	Document dependencies
Data Sharing	partner Snowflake account	Export via open formats
Snowpipe	proprietary ingestion	Replace with Airbyte or COPY INTO

MIGRATION COST IF YOU LEAVE

- Rebuild all ingestion pipelines
- Recreate security policies & role hierarchy
- Re-implement Time Travel logic elsewhere
- Export all data — egress costs apply

Real Risk for SME

Proprietary features (Time Travel, Snowpipe, Zero-Copy Clone) are easy to adopt but create silent dependencies that make migration costly if leaving snowflake.

Mitigation Strategy

Prefer ANSI SQL + open formats where possible. Use Snowflake-specific features consciously, not by default.

Cost Scenario: SwissBike SA

ASSUMPTIONS

- AWS US East (N. Virginia), Standard edition
- X-Small **Gen 2** warehouse → 1.35 credits/h
- 10h/week → 43.33h/month
- 1 TB storage + CoWork + Cortex AI enabled

$$10\text{h/week} * (52 \text{ weeks} \div 12 \text{ months}) = 43.33\text{h/month}$$

HOW SNOWFLAKE CHARGES

LAYER	RATE	CHARGED WHEN
Compute	\$2.00/credit	Warehouse running only
Storage	\$23.00/TB	Always, 24/7
Cloud Services	free	< 10% of compute

FORMULA

$$43.33\text{h} \times 1.35 \text{ cr/h} \times \$2.00/\text{cr} = \$117.00/\text{month}$$

MONTHLY BILL

COMPONENT	COST
Compute (XS Gen 2)	\$117.00
Snowflake CoWork	\$123.08
Snowflake Cortex AI	\$5.78
Storage (1 TB)	\$23.00
Total	\$268.86

Key cost controls

- **AI features** - CoWork + Cortex AI cost **\$128.86/month** more than compute itself. Enable consciously.
- **Auto-suspend** : without it, ~\$194/month compute vs \$117
- **60s minimum** : avoid frequent short sessions
- **Storage billed on compressed size** — auto columnar compression saves cost
- **Ingress free but egress charged** cross-region (\$20/TB v.s.\$90/TB)

Pros & Cons

	ADVANTAGE	TRADE OFF
Architecture	Decoupled compute & storage	Cold start 2–3s
	MPP – concurrent users, no interference	Not for OLTP
Compute	Pay per second, zero when suspended	No hard spend cap by default
Storage	Columnar compression – 3–5x automatic	\$26.95/TB – most expensive in EU
	Pruning – query speed constant as data grows	Egress costs on data extraction
	Time Travel + Fail-safe – 90d + 7d recovery	No free tier – trial only
Vendor	Multi-cloud – AWS, Azure, GCP	Proprietary lock-in – Time Travel, Snowpipe
	ANSI SQL – portable	AI features – 46% of bill
Other	Encryption, RBAC, SSO out of the box	SQL only – no low-code option

Snowflake vs Market Alternatives

	SNOWFLAKE	BIGQUERY	REDSHIFT	AZURE SYNAPSE
Owner	Independent	Google	Amazon	Microsoft
Setup complexity	Low	Lowest	Medium	High
columnar + auto compression	yes	yes	yes	yes
Scaling	Manual config required for verticle scaling	Fully automatic	Manual	Semi-automatic
Lock-in	Proprietary features	GCP ecosystem	AWS ecosystem	Microsoft ecosystem
Pricing	Credits + storage	Per TB scanned	Per node 24/7	Per DWU + storage
Storage/TB (EU)	\$26.95 — most expensive	~\$20	~\$24 bundled in node	~\$20
Free tier	\$400 trial, first 30 days	1TB queries + 10GB/month	X	X

All 4 have vendor lock-in. BigQuery is simpler and cheaper for smaller workloads. Snowflake wins when **cloud neutrality and multi-source integration matter.**

Recommendation

USE SNOWFLAKE WHEN

- ✓ SMEs needing centralized analytics
- ✓ Data volume growing beyond single database capacity
- ✓ Organizations with multiple teams accessing data
- ✓ Businesses seeking low operational overhead
- ✓ Multiple disconnected systems need centralizing
- ✓ no existing AWS/Azure/GCP commitment

NOT RECOMMENDED FOR

- ✗ Small transactional applications (OLTP)
- ✗ Organizations requiring full infrastructure control
- ✗ Businesses with very limited cloud adoption
- ✗ Already on AWS/ Google → Redshift / BigQuery
- ✗ Budget too small – BigQuery free tier covers basic SME needs
- ✗ Need real-time or transactional workloads (OLTP)
- ✗ No SQL knowledge

Final Assessment

Choose Snowflake when: cloud-agnostic, multi-source data, growing volume, no cloud vendor preference.

Otherwise: BigQuery (GCP), Redshift (AWS), Synapse (Azure) are cheaper native alternatives.

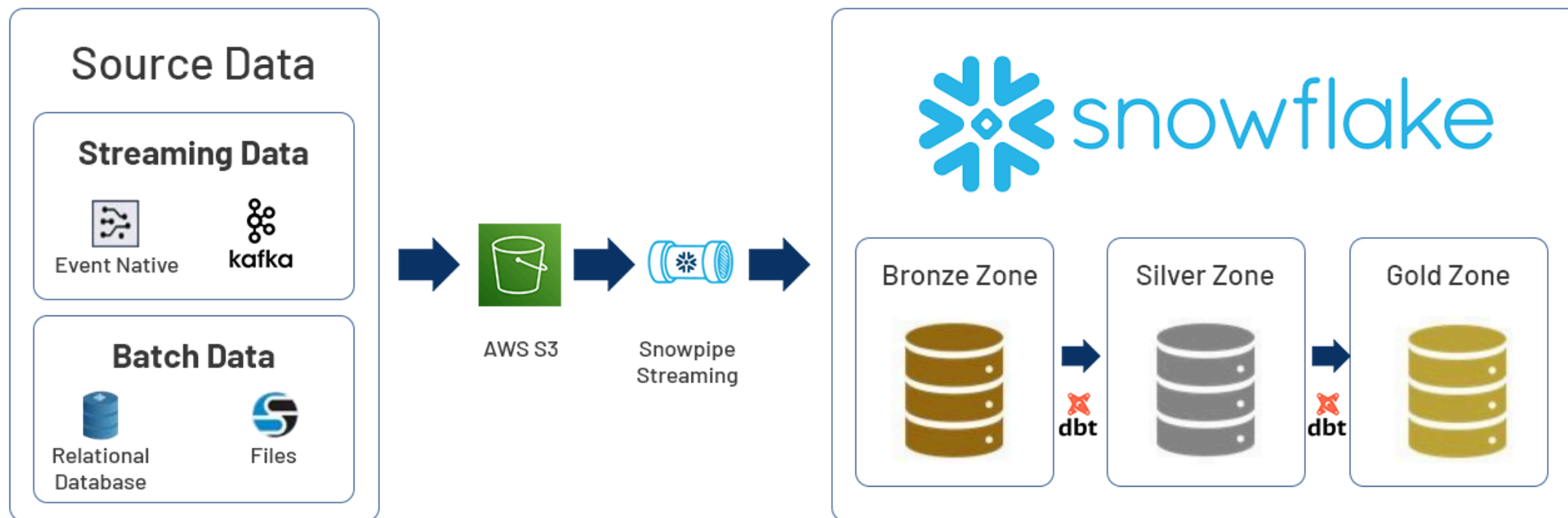
Demo

References

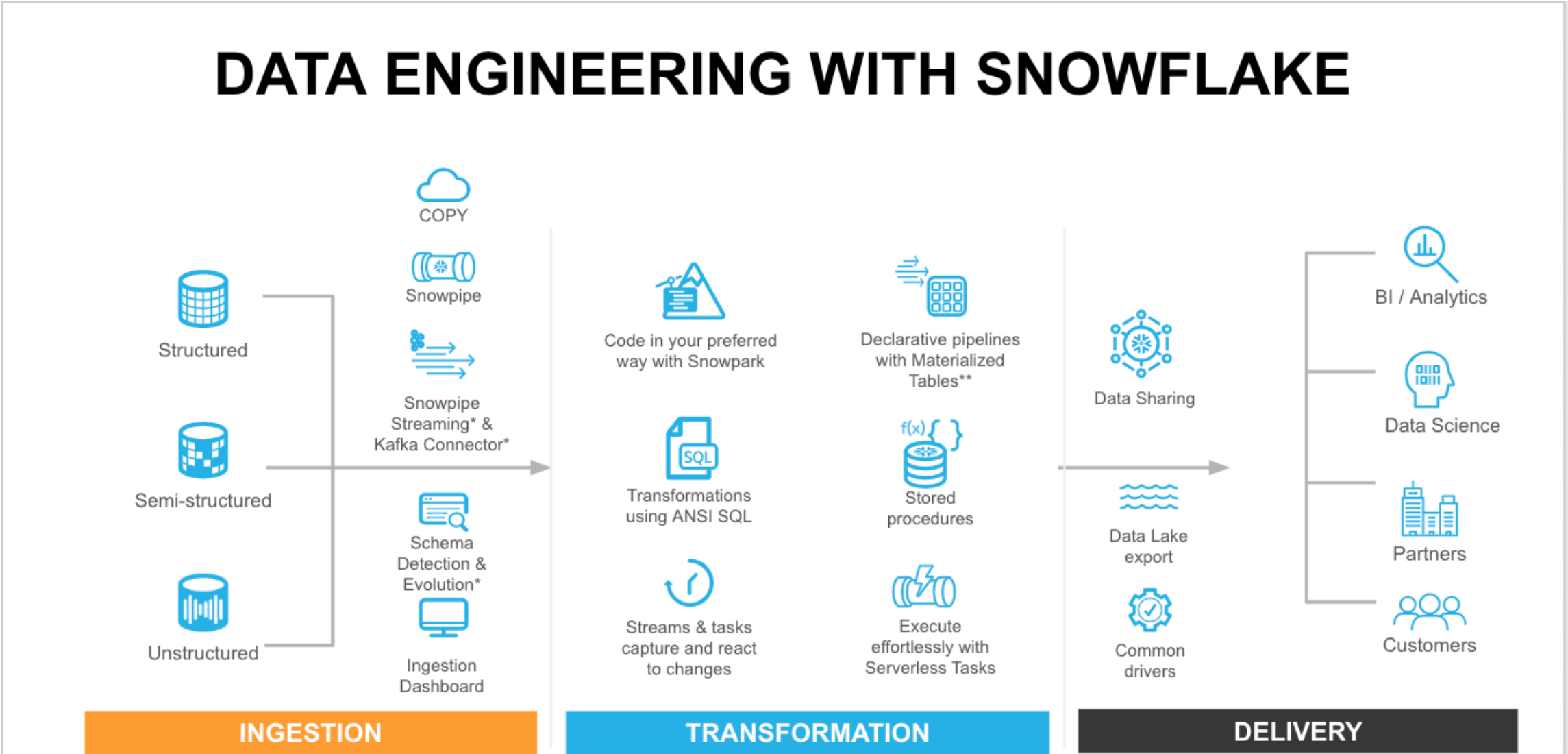
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#Below diagrams are only used for Q&A if needed

Data Zone



Data Pipeline



* in preview
** in development

Data sharing

